

QR Codes To Transmit A Submersible's Position

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Due to the rapidly changing and contested warfare environment, organizations are seeking to diminish their use of IR means of communication due to an increase in the ability to destroy or jam IR data transmission. The intention of this paper is to propose the use of QR codes on the back of drones in order for EO satellites to be able to read data, removing the necessity of IR.

QR codes are excellent ways to transfer information via electro-optics and operate in a simple manner. Within a QR code, there are three squares in each corner that anchor the orientation of the QR, this allows the software to understand where to begin, and where to conclude reading. Unlike bar codes, which only function on one axis, QR codes encode information on two axes allowing for a dramatic increase in the amount of information stored (Ruoti). QR codes store information commonly via black or white squares, which each square representing a 1 or a 0.

Due to the varying information our drone would read, it is only reasonable to attempt to create a QR code that can vary its information being relayed. In order to accomplish this, the black and white squares on a standard QR must be able to change its state. We propose that lights be used in place of the static black and white squares in order to entirely change the information transmitted by the QR code. Initially, we propose a 15 x 15 grid of lights. This would allow for a 5x5 square in three corners to be static in color and act as our anchoring points. The remaining free use lights would then allow for 1.425×10^{45} individual combinations of information to be transferred.

You may ask, why not a singular light that flashes on and off allowing us to reduce the clarity at range out camera would require? Although that may seem the most simple solution, we do not want to create a initial limitation in relation to the amount of data transmitted incase this project is further expanded upon. However, our group would need to experiment with the optimal spacing and amount of lights that would allow for the highest image resolution for the satellites.

Due to the distance between the drone and the satellite, we need to optimize either the color the lights project, and or pick the best color space to process. This would allow for the best image resolution at a great distance. While it is easy to make the lights project neon-orange and or yellow (the best colors for ocean background contrast), we should consider the color scale in which the computer perceives images and process differences pixel intensities. Historically, such as in lane detection, the difference between pixel colors can be determined using thresholds. This threshold uses the ability to detect the difference in pixel intensities in order to tell the difference between objects (Storsaeter). Commonly, RGB (red, blue, and green) and gray scale are used in cameras for detection; however, research was done and found that HSV, HSL, and CEILAB were the best color scales to detect lane lines in sub-optimal conditions such as snow covered roads and I believe this logic could be applied to our project due to sub optimal conditions, such as sea state. Thus, increasing the resolution when observing out QR code.

Although this proposal seeks to rid of IR communication in order to reduce jamming; however, there could be potentially security risks due the use of QR codes. QR codes can be spoofed and used to hide malicious information (Ruoti). Thus I propose we use a unique key within the information being relayed in order to confirm the validity of our message and prevent

a malicious actor flying drones out with a message that may distort the real-time presence of an underwater submersible.

Ruoti, Scott. "How QR codes work and what makes them dangerous - a computer scientist explains." NextGov.com, 13 Dec. 2023. Gale Academic OneFile, link.gale.com/apps/doc/A819489937/AONE?u=anna82201&sid=bookmark-AONE&xid=c578784f. Accessed 15 Sept. 2026.

Storsæter, Ane Dalsnes, et al. "Camera-Based Lane Detection—Can Yellow Road Markings Facilitate Automated Driving in Snow?" *Vehicles*, vol. 3, no. 4, Oct. 2021, pp. 661–90, <https://doi.org/10.3390/vehicles3040040>.